

Utility, Model Differences, and Code Experiment

Fertility and Housing Supply

Fertility and Housing Supply Project

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Purpose

This document gives a detailed model explanation focused on what changes relative to the baseline NIMBY paper, why those changes matter, and what was learned from a first coding experiment.

1 Baseline NIMBY model block (reference point)

The original baseline (as used in the NIMBY project) can be summarized as:

1. Household utility over consumption and housing services (CRRA/Cobb-Douglas).
2. Life-cycle decision problem for tenure, assets, housing, and consumption.
3. Housing market/political equilibrium where supply outcomes (prices/quantities) are shaped by voting incentives.
4. Exogenous demography in transition calculations.

The fertility project keeps (1)–(3) as much as possible, but changes (4) and links household fertility choices to housing and political dynamics.

2 Exact differences from the NIMBY paper

Difference A: Utility now includes children-at-home and crowding

Baseline flow utility:

$$u(c_t, h_t) = \frac{(c_t^{1-\zeta} h_t^\zeta)^{1-\gamma}}{1-\gamma}.$$

Fertility extension flow utility:

$$u_t = \frac{(c_t^{1-\zeta} h_{t,eff}^\zeta)^{1-\gamma}}{1-\gamma} + \chi \frac{n_{t,home}^{1-\eta}}{1-\eta}, \quad h_{t,eff} = \frac{h_t}{(1 + \lambda n_{t,home})^\psi}.$$

Economic meaning:

- Housing enters utility through *effective* space, so crowding makes the same dwelling less valuable for larger families.
- Fertility is not only a budget burden; it can generate direct utility via the child term.
- This creates a quantity–space tradeoff even before adding child quality spending.

Difference B: Fertility is now a household choice variable

In fertile ages, births are chosen jointly with consumption/housing/saving:

$$V_t(s_t) = \max_{c_t, a_{t+1}, h_{t+1}, b_t} \{u_t + \beta E_t V_{t+1}(s_{t+1})\}$$

subject to the extended budget:

$$c_t + q_t^h h_{t+1} + a_{t+1} + \kappa(b_t, n_{t,home}) \leq y_t + (1 + r_t)a_t + \text{housing cash-flow terms.}$$

Economic meaning:

- In the baseline, demographics do not respond to housing affordability.
- In the extension, higher housing costs can delay or reduce births through both utility crowding and budget channels.

Difference C: Children leave home explicitly

Children-at-home state evolves with cohort aging and a leave-home rule (deterministic benchmark):

$$\begin{aligned} N_{0,t+1} &= b_t, \\ N_{a+1,t+1} &= N_{a,t}, \\ n_{t,home} &= \sum_{a=0}^{A_{leave}-1} N_{a,t}. \end{aligned}$$

Economic meaning:

- Utility and housing demand respond to *dependent* children, not lifetime births.
- This allows realistic life-cycle housing demand reversals (space pressure rises then later falls).

Difference D: Demography becomes endogenous

Cohort masses evolve with endogenous births:

$$\begin{aligned} M_{j+1,t+1} &= \ell_j M_{j,t}, \\ M_{1,t+1} &= \sum_{j \in \mathcal{F}} f_j M_{j,t}. \end{aligned}$$

Economic meaning:

- Fertility decisions alter future age distribution.
- Age distribution alters future median-voter composition.
- Median-voter composition alters future supply outcomes.

Difference E: Equilibrium is now a joint price–demography fixed point

Instead of solving prices/policies with exogenous demographic weights, the extension solves:

$$\{p_t, r_t, \text{policy}_t, M_t\}_{t \geq 0}$$

jointly, because each object feeds back into the others over time.

3 Why this matters for your paper contribution

The key conceptual gain is that the model now has a full mechanism from housing supply restrictions to fertility outcomes:

planning/supply $\rightarrow$ housing prices + unit mix $\rightarrow$ crowding + fertility choice $\rightarrow$ future demography.

This is stronger than treating fertility as an exogenous reduced-form trend because it ties fertility outcomes to policy-relevant housing margins.

4 Code experiment: objective and design

Objective

The prototype was not intended for calibration or quantitative policy claims. Its purpose was to test numerical behavior of the new fixed-point structure and confirm whether the convergence concerns in the model setup are operationally important.

Prototype architecture

A compact simulation was implemented with:

- age shares (8 cohorts),
- endogenous births in fertile cohorts,
- a political-tightness index increasing with older-share and decreasing with younger-share,
- price updates from demand–supply imbalance,
- optional family-sized supply shock,
- outer-loop damping parameter.

Scenarios run

1. No damping (aggressive update).
2. Damping $\xi = 0.25$ (stabilized update).
3. Damping $\xi = 0.25$ plus family-sized supply shift from period 15.

5 Code experiment results and diagnostics

Main diagnostics

Scenario	Avg price (t=10..49)	Avg fertility (t=10..49)	Max $ \Delta p_t $
No damping	-5.939	0.723	2.426
Damping 0.25	-2.597	0.416	1.948
Damping 0.25 + policy	-2.638	0.443	1.349

Interpretation:

- Undamped updates generate large jumps and near-corner behavior.
- Damping reduces oscillation and keeps trajectories in a more interpretable range.
- In the damped setting, family-sized supply support increases fertility and lowers transition volatility.

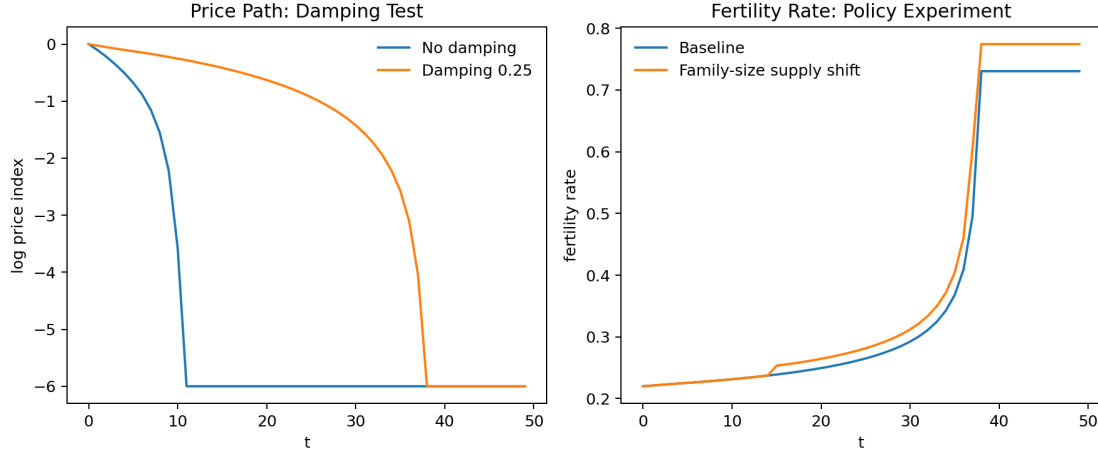


Figure 1: Prototype dynamics under alternative update rules and policy shift

Failure log (important)

Early runs produced exponential overflow and pathological paths. Two stabilization changes were required:

1. Bounding exponent arguments in demand/supply transforms.
2. Clamping price updates and using damped outer-loop updates.

This is direct evidence that the full MATLAB implementation should start with tight numerical guards.

6 Implementation implications for the full model

What should be coded first

1. Household block with fertility control and children-at-home state.
2. Demographic transition block that consumes policy functions and updates cohort masses.
3. Outer fixed-point wrapper with damping and share normalization.

Recommended convergence checks

Use three tolerances, not one:

- price-path distance,

- cohort-share distance,
- aggregate births distance.

All three should converge before declaring equilibrium solved.

Suggested baseline robustness grid

- Damping values: $\xi \in \{0.10, 0.25, 0.40\}$.
- Leave-home age: $A_{leave} \in \{18, 19, 21\}$.
- Child-cost curvature and crowding intensity: low/medium/high triplets.

Bottom line

The fertility extension is not a marginal tweak to the NIMBY model; it changes the equilibrium object from a housing-policy fixed point with exogenous demographics to a joint housing–demography fixed point. The prototype confirms that this is computationally feasible but numerically delicate, so disciplined damping and normalization should be treated as core model ingredients.